

JavaScript Learning Summary

Covered Topics

1. ECMAScript – JavaScript follows the ECMAScript specification.
2. ECMAScript Versions – ES5, ES6, ES2017, ES2020 and major features.
3. JIT Compiler – Optimizes frequently executed (hot) code.
4. JIT De-optimization – Falls back when assumptions become invalid.
5. Garbage Collection – Automatic memory cleanup.
6. Mark and Sweep – Reachable objects survive, unreachable objects are removed.
7. Symbol – Unique primitive used to avoid property-name collisions.
8. Error Object Properties – name, message, stack.
9. Custom Errors – Creating application-specific error types.
10. AggregateError – Represents multiple errors together.
11. Nullish Assignment (??=) – Assigns only when value is null/undefined.
12. structuredClone() – Deep cloning utility for data structures.
13. Shallow Copy vs Deep Copy – Reference sharing vs full duplication.
14. eval() – Executes strings as JavaScript code (generally avoided).
15. Labels – Named loops/blocks used with break and continue.
16. Date Handling – Date creation and common methods.
17. Parent of Array – Array.prototype.
18. Map Prototype – Map instances inherit from Map.prototype.
19. bind() Return Type – Returns a new function.

Key Examples

Symbol

```
const id = Symbol("id");  
obj[id] = 123;
```

Garbage Collection

```
let user = {name:"Ameen"};  
user = null; // object becomes collectible if no references remain
```

Shallow Copy

```
const copy = {...obj};  
Nested objects remain shared.
```

Deep Copy

```
const copy = structuredClone(obj);  
Nested objects become independent.
```

bind()

```
const bound = greet.bind(user);  
typeof bound === "function";
```

Pending Topics

1. DevTools Debugging Basics
2. JavaScript Coding Conventions
3. Side Effects
4. Memory Leaks

Main Roadmap

Closures
Hoisting
TDZ (Temporal Dead Zone)
Lexical Environment
Scope Chain
this Keyword
call(), apply(), bind()
Event Loop
Microtask Queue
Callback Queue
Promises
Async/Await
Promise Methods
Prototype
Prototype Chaining
Prototypal Inheritance
Classes
Encapsulation
Abstraction
Polymorphism
Debouncing
Throttling
Web Workers
Proxy
WeakMap
WeakSet
DOM
Event Bubbling
Event Capturing
Event Delegation
Local Storage
Session Storage